INCH-POUND
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MILITARY SPECIFICATION

HUMIDITY SENSOR, RESISTANCE

This specification is approved for use by Communications-Electronics Command, Fort Monmouth, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of humidity-sensitive electrical resistance sensor (hygristor) which is used for measuring atmospheric relative humidity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following secifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-380	-Plastic Molding Material, Methacrylate.
L-P-396	-Plastic Molding, Polystyrene.

MILITARY

MIL-P-116	-Preservation, Methods of.
MIL-P-11268	-Parts, Materials and Processes used in Electronic Communication Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Communications-Electronics Command; ATTN: AMSEL-ED-TC, Fort Monmouth, New Jersey 07703-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC-N/A FSC 6685

DISTRIBUTION STATEMENT A. Approved for public release; distribution unlimited.

MIL-C-45662	-Calibration of Standards.
MIL-R-49440	-Radiosonde Set ML-659(V), Family of.
MIL-E-55585	-Electronics Equipment and Parts, Packaging Of.

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252	-Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices.
MIL-STD-454	-Standard General Requirements for Electronic Equipment.
MIL-STD-726	-Packaging Requirements Code

2.1.2 Other publications. The following documents of the issue in effect on date of invitation for bids form a part of this specification.

U.S. WEATHER BUREAU

W.B. #235	Psychrometric Table.
W.B. #1071	Psychrometric Table (With Supplementary Tables #31).

(Copies of U.S. Weather Bureau Publications may be obtained from Superintendent of Documents, Government Printing Office, Washington 25, D.C.).

SMITHSONIAN INSTITUTE

Smithsonian Meteorological Tables
(Sixth Edition Revised)

(Copies may be oibtained from Smithsonian Institute, Washington 25, D.C.). (Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) the contractor shall furnish first article units for first article inspection and approval (see 4.3).

3.2 Construction. The Humidity Sensor, Resistance shall consist of a plastic plate to which electrodes are attached. A hygroscopic coating containing carbon, humeoctant, platicizer and binder shall be evenly disposed between the electrodes and cover an area of the base sufficient to adjust resistance at 25°C and 33 pecent RH to the value specified in 3.9.1. The film of carbon provides variable electrical resistance with variations in moisture environment over the relative humidity range of 10 to 100 percent and operating temperature range of +40°C to -40°C. The variation in resistance shall be calibrated in terms of relative humidity to provide relative humidity indications.

3.2.1 Electrode. The electrodes shall be continuously mechanically bonded along the longitudinal edges of the plastic plate. The electrodes shall also make electrical contact along their entire length with the humeoctant film on both sides of plastic plate. The electrodes shall contain no cracks.

3.2.2 Dimensions. The physical dimensions of the plastic plate shall be as follows:

Length (in.)	Width 1/ (in.)	Thickness (in.)
2-1/2 ± 1/16	11/16 ± 1/32	0.04 ± 0.01 (not including electrode thickness)
(6.3 ± 1.6mm)	(17.5 ± 0.8mm)	(1.02 ± 0.25mm)

1/ The minimum width of each electrode shall be 3/32 inch (2.4mm).

3.2.3 Warpage. Longitudinal or transverse warpage shall not exceed 1/16 inch (1.6mm).

3.2.4 Container. The container shall be a hermetically sealed metal can. The maximum external dimensions of the sensor and internal packing materials. One inexpensive can opening device shall be supplied with each 24 Humidity Sensors.

3.3 Abbreviations and definitions. The following abbreviations or definitions used throughout this specification are as defined below:

3.3.1 Relative humidity. Relative humidity shall be abbreviated RH and shall mean the amount of moisture in the air as compared with the amount that the air could contain at the same temperature, expressed as a percentage. All values herein are expressed with reference to U.S. Weather Bureau psychrometric tables and are based on extrapolated values for subcooled water below freezing conditions.

3.3.2 Humidity. Humidity, in this specification, shall mean relative humidity and shall be expressed in percent as outlined in 3.3.1.

3.3.3 Symbols. The following symbols shall represent the quantities indicated wherever they are referenced herein:

(a)	R_{33}	Resistance of an element at a particular temperature and 33% RH, when the 33% RH level is approached with ascending humidity conditions.
(b)	^{o}C	Degrees centigrade (Celsius).
(c)	mb	Millibars air pressure.
(d)	R	Resistance of an element measured in accordance with 3.9.
(3)	R/R_{33}	Resistance ratio for a particular sensor determined by dividing the resistance of (d) above by the resistance of (a) above for that sensor.

3.4 Parts, materials and processes: general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein shall apply. (see 4.4).

3.4.1 Plastic parts. The surfaces of all plastic parts shall be clean and free from dust, dirt, grease or other surface impurities prior to coating and all edges shall be free from burrs or protrusions resulting from cutting, sawing, punching or other machining operations. Plastic material shall conform to 3.4.2.

3.4.2 Molded thermoplastic. Molded thermoplastic material, where used, shall be polystyrene in accordance with L-P-396, or methacrylate in accordance with L-P-380.

3.4.3 Cleaning. After metal parts have been fabricated they shall be cleaned in accordance with best commercial practices. The cleaning process shall have no deleterious effect on the equipment. All traces of corrosive compound shall be removed to prevent subsequent corrosion of parts themselves or other parts adjacent thereto.

3.4.4 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.5 Marking. To each sensor container shall be attached a printed label. With the manufactuer's name, part number, date of manufacture, lot number and sensor lock-in resistance value (see 4.4).

3.6 Safety Engineering.

3.6.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.11).

3.6.2 Corner and edge rounding. Longitudinal edges of the plastic plate shall be acceptable as extruded with rounded edges. Transverse edges shall be free of gross defects and subject to requirements of MIL-STD-454 regarding personal safety.

3.6.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.7 Weight. The weight of the element shall not exceed 10 grams.

3.8 Interchangeability. All Humidity Sensors shall be mechanically and electrically interchangeable. (see 4.7)

3.9 Performance requirements. The Humidity Sensor shall meet the following performance requirements.

3.9.1 Resistance at 33% R.H. and 25°C. The Resistance at 33% R.H. shall be 10,000 ±5000 ohms. The data shall be provided with each sensor with an accuracy equivalent to 5 percent R.H.

3.9.2 Resistance-Humidity transfer characteristic. The resistance humidity ratio at 25°C shall be defined as the ratio of the resistance of the sensor at any humidity to the resistance at 33% R.H. The humidity sensor resistance vs humidity transfer characteristic is described by Equation 1. The supplied coefficients shall be applicable for all humidity sensors furnished in the applicable contract. The resisance of the humidity sensor shall conform to this equation with an accuracy equivalent to ±5 percent R.H.

Equation 1.

$$RH_W = A - \left[\frac{69}{\sum_{K=0}^{4} D_k \left(f\ (R_S)\right)^K} \right]$$

RH_W = The percent relative humidity with respect to water.

$f(R_S)$ = $f(t)\ln(R_S/R_{33})$; $f(t)$= 1.00 at +25° C.

R_S = The humidity sensor resistance (the effect of the 1.2 megohm resistor must be compensated for when computing the sensor resistance).

R_{33} = The humidity sensor resistance at 33 percent RH at +25°C.

Where $f(R_S) \geq -.2$	Where $f(R_S) < -.2$
D0 = 1.000 E+00	D0 =-2.440 E+00
D1 = 7.290 E-01	D1 =-7.370 E+00
D2 = 5.580 E-02	D2 =-9.980 E+01
D3 = 7.480 E-03	D3 =-2.514 E+02
D4 = 1.010 E-02	D4 =-2.110 E=02
A = 102	A = 0

3.9.3 Temperature - humidity transfer characteristic. The humidity sensor temperature characteristic $f(t)$, is described by Equation 2. The supplied coeficients shall be applicable for all humidity sensors furnished in the contract. The effect of temperature shall be kept to minimum. The humidity sensor shall conform to Equation 1 with an accuracy equivalent to ±5 percent R.H.

Equation 2. $$f(t)=\sum_{k=0}^{3} C_k(t)^K$$

Where: $f(t)$= temperature coefficient of the humidity sensor

t = temperature in degrees Celsius.

$R_S/R_{33} \geq 1{:}00$	$R_S/R_{33} < 1.00$
C_0 = 7.885 E-01	C_0 = 9.243 E-01
C_1 = 9.286 E-03	C_1 = 3.059 E-03
C_2 = -2.462 E-5	C_2 =-1.188 E-06
C_3 = -3.368 E-07	C_3 = 0

3.9.4 Hysteresis. The sensor shall be manufactured to minimize any change in the resistance humidity transfer characteristic for measurements under decreasing humidity conditions, as opposed to measurements made under increasing humidity conditions. Under decreasing humidity conditions, the resistance humidity transfer characteristic allowance (3.9.2) shall be increased by 3 percent R.H.

3.9.5 High humidity operation. The sensor shall exhibit a continuously increasing resistance when tested under increasing humidity conditions.

3.9.6 Repeatability. Any sensor that has been subjected to any specified performance tests, except for the high humidity test of 4.8.6 and the low and high temperatures of 4.8.4 shall remain within the specified tolerance of calibration accuracy when subjected to recheck at +25°C of 4.8.2 within 24 hours of having been subjected to the former tests (see 4.8.7).

3.9.7 Nonintersection of characteristic curves. The RH required to maintain any constant R/R_{33} resistance ratio greater than that obtained at 50 percent RH and +25°C., shall decrease as the temperature is decreased over the range of +40°C to -40°C., inclusive. Compliance shall be determined by examination of the plots of the resistance ratio vs percent RH characteristic curves for constant temperatures over the operation range as determined in 4.8.8.

3.9.8 Time constant. Sixty-three percent of the total change in resistance of the element shall occur in less than 2 seconds at +25°C and in less than two minutes at -30°C when subjected to the following conditions.

a.	Air velocity:	Not to exceed 1000 ft/min
b.	Atmospheric pressure:	960 to 1,040 millibars
c.	Initial humidity:	20 + 10, -5 percent R.H.
d.	Final humidity:	80 + 10, -5 percent R.H.

3.10 Service conditions. The humidity Sensor shall meet the following nonoperating service conditions (see 4.9.).

3.10.1 Temperature humidity (storage). The sensor shall meet the requirements of 3.9 after subjection to the following nonoperating conditions:

a. Any ambient temperature ranging from +60°C to -60°C.
b. Any environment relative humidity in the range between zero and 100 percent. There shall be no impairment of the physical or electrical characteristics of the sensor beyond the limits specified herein by storage for period up to three years in its hermetically sealed container at any probable combination of temperature and humidity.

3.10.2 Altitude. The sensor in its sealed container shall meet the requirements of 3.9 after subjection to any altitude up to and including 50,000 feet (125+5mb) above sea level.

3.10.3 Vibration. The sensor in its sealed container shall withstand low frequency vibrations as encountered in transportation and storage.

3.11 Aging. All sensors shall be subjected to an aging period of 60 days (minimum) prior to government inspection and shipment. The dipping date of sensor shall be clearly marked on the label. (see 4.10).

3.12 Workmanship. The Humidity Sensor shall be manufactured and assembled in accordance with Requirements 5, 7 and 9 of MIL-STD-454 and the applicable portions of the following paragraphs of MIL-STD-11268 (see 4.11):

3.2	Construction
3.2.1	Electrodes
3.3.3	Dimensions
3.2.4	Container
3.4	Parts, materials and processes, general
3.4.1	Plastic parts
3.4.3	Cleaning
3.8	Interchangeability

...constructed in a repair shop environment

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspections. The inspection requiremens specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspection of equipment before packaging (see 4.5).
d. Packaging inspection (see 4.12).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish 30 first article units of the humidity sensors for group A, B and C testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table I, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test reports shall be as required in the contract.

TABLE I. First article inspection.

Inspections	Reqt para	Test para	Order of tests 30 units
1. Inspections covered by subsidiary documents 1/	3.4 3.5	4.4	Inspection to be performed on all units.
2. Group A inspection 1/	See table II		Test to be performed on all units.
3. Group B inspection 1/	See table III		Tests to be performed on all units.
4. Group C inspection Temperature Humidity Altitude Vibration	 3.10.1 3.10.1 3.10.2 3.10.3	 4.9.1 4.9.2 4.9.3 4.9.4	 3 4 2 1

1/ The inspection 1 to 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

Item	Where required
Parts, materials, and processes; general	3.3
Marking	3.6

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspections which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspection in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection level III and AQLs indicated in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL major	AQL minor
Visual and mechanical 1/	3.12	4.11	1%	4%
System safety	3.6	4.11	1%	4%
Performance 1/	3.9	4.8	4%	10%
Resistance at 33%	3.9.1	4.8.3	for the	for the
Resistance-humidity transfer characterist.	3.9.2	4.8.2	group	group
Temperature-humidity transfer charact.	3.9.3	4.8.4		
Hysteresis	3.9.4	4.8.5		
High humidity operation	3.9.5	4.8.6		

1/ See 4.5.1.2

4.5.1.2 Classification of defects. The defects of group A inspection shall be classified as follows:

Visual and mechanical inspection

Can seal
Major - Can seal leaks
Minor - N/A

Performance

Baseline accuracy
Major - Resistance error exceeds equivalent humidity by more than 7 percent R.H.
Minor - Resistance error exceeds equivalent humidity by more than 5 percent R.H.

Calibration accuracy
Major - Error exceeds limits by more than 2 percent R.H.
Minor - Error exceeds limits, but by not more than 2 percent R.H.

Hysteresis
Major - Exceeds limits by more than 3 percent R.H.
Minor - Exceeds limits by not more than 3 percent R.H.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B sample plans, for the AQL's listed in table III, shall be inspection level 5-4.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL
Weight	3.7	4.6	6.5%
Interchangeability	3.8	4.7	6.5%
Repeatability	3.9.6	4.8.7	6.5%
Nonintersection of characterisic curves	3.9.7	4.8.8	6.5%
Time constant 1/	3.9.8	4.8.9	6.5%

1/ The test performed at first article inspection and at the beginning of each lot production. A lot is the quantity fabricated from a single batch of carbon film solution.

4.5.3 Group C inspection. Group C inspection shall be performed on inspection lots that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection. Thirty Humidity Sensors shall be selected at random from each lot produced. A lot is the quantity fabricated from a single batch of carbon film solution.

4.5.3.2 Group C failures. Action required relative to group C failures shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. Group C testing is considered to be destructive. The same sample element shall not be to more than one of the tests specified by group C testing. All samples subjected to group C testing shall be discarded.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Temperature	3.10.1	4.9.1
Humidity	3.10.1	4.9.2
Altitude	3.10.2	4.9.3
Vibration	3.10.3	4.9.4

4.6 Weight. The Humidity Sensor shall be weighed on a calibrated scale to determine compliance with weight requirement of 3.7.

4.7 Interchangeability. The dimensions listed in 3.2.2 shall be measured to determine compliance with the physical interchangeability requirements of 3.8 (see 4.11).

4.8 Performance testing. The Humidity Sensor shall be tested for compliance with specified performance requirements of 3.9. The measurements shall be performed in accordance with established inspection standard specified in 4.8.1.

4.8.1 Inspection standards. Test equipment shall conform to the requirements of MIL-C-45662.

4.8.1.1 Resistance measuring instrument. All resistance measurements shall be made with a 60 Hz ac conductivity bridge. The current through the sensor under test shall not exceed 23 ma. The instruments used shall have a calibration accuracy of ±0.1%. A digital voltmeter with accuracy of ±0.1 percent or better may be substituted.

4.8.1.2 Relative humidity standard for 25°C measurements. The primary standard for determination of relative humidity shallk be a wet and dry bulb psychrometer, using matched precision thermometers (±0.1°C certified accuracy) and approved wicking and distilled water, the two pressure humidity generators described in "Pressure and Humidity Apparatus" referenced below or equivalent, air flow shall be in accordance with instructions and tables of the U.S. Weather Bureau Psychemetric Tables WB No. 235 and WB No. 1017 or Pressure and Humidity Apparatus -Journal of Research of the National Bureau of Standards V48 p.269 (1952). At temperatures below 0°C, the two pressure humidity generator shall be used.

4.8.1.3 Humidity calibration chamber. A humidity chamber shall be provided in which the relative humidity and temperature can be maintained constant within the following limitations:

	Temp.	%RH
a. At temperature from +40°C to 0°C inclusive	±0.2°C	±1%RH
b. At temperature below 0°C	±0.2°C	±2%RH

The temperature at all points within the working space more than two inches from chamber walls shall be uniform within 0.5°C.

4.8.1.4 Standards for +40°C., 0°C., -20°C, and -40°C, calibration test. Dry bulb temperature shall be measured by means of a standard thermometer or equivalent having a certified accuracy of ±0.1°C. Relative humidity shall be determined by National Bureau of Standards two-pressure humidity generator or equivalent as in 4.8.1.2.

4.8.1.5 Tables. Determination of relative humidity shall be made from Smithsonian Meterological Tables, latest revised edition, utilizing the readings of dry bulb dew point temperature taken from the standards in 4.8.1.4 when the humidity chamber is maintained at ±0.5°C dew point temperature, or calculated as referenced in 4.8.1.2.

4.8.2 Resistance-humidity transfer characteristic. The humidity sensor shall be placed in a test chamber at 25 ±2°C.

Resistance measurements shall be made at each of the following humidity points in the order shown.

a. 15% ±5%
b. 33% ±2%
c. 50% ±5%
d. 70% ±5%
e. 90% ±5%

The indicated humidity shall be determined using Eqn. 1. The difference between this indicated humidity and the chamber humidity is the transfer characteristic accuracy. The accuracy shall conform to the requirement of 3.9.2.

4.8.3 Resistance at 33% R.H. and 25°C. The resistance measured at 33% R.H. and 25°C shall conform to the requirements of 3.9.1.

4.8.4 Temperature humidity transfer characteristic. The resistance of the sensor shall be measured at a temperature of -35 ± 5°C. The indicated humidity shall be determined using Equation 2. The difference between this indicated humidity and the chamber humidity shall conform to the requirements of 3.9.3.

4.8.5 Hysteresis. The sensor shall be placed in a test chamber at 25 ± 2°C. Resistance measurements shall be made at the following humidity points in the order shown.

a. 90% ±5%
b. 70% ±5%
c. 50% ±5%
d. 33% ±5%
e. 15% ±5%

The indicated humidity shall be determined using Eqn. 1. The sensor shall conform to the requirements of 3.9.4.

4.8.6 High humidity operation. The humidity sensor shall be placed in a test chamber at 25 ± 2°C. Resistance measurements shall be made at each of the following humidity points, only in ascending direction.

a. 90% ±1%
b. 92% ±1%
c. 94% ±1%
d. 96% ±1%

The resistance values shall conform to the requirements of 3.9.5.

4.8.7 Repeatability test. Within 24 hours after having been subjected to specified performance tests, except high humidity test of 4.8.6 and the low temperature of 4.8.4, the sensor shall be subjected to 4.8.2 test at 25°C to determine compliance with the requirements of 3.9.6.

4.8.8 Nonintersection of characteristic curves. Compliance shall be determined from an examination of a plot of resistance ratio (ordinate) versus percent R.H. (abscissa) characteristic curves for constant temperatures as obtained in 4.8.4. Any evidence of intersection of characteristic curves in any of the samples tested shall constitute cause for rejection of the samples (See 3.9.7).

4.8.9 Time constant test. The entire test shall be conducted at +25°C ±2°C. The sensor shall be placed in a chamber and electrically connected in series with a +0.1 volt ac source, a 500 microampere ac meter and subjected to a stable R.H. of 20 +10, -5 percent. The circuit shall be closed 30 seconds before making the first observation of the meter reading. This shall be considered the reading at "zero time". At this time the conditions shall be quickly changed in such a way that the element is exposed to an air mass having a stable R.H. of 80 +10,-5 percent and an air velocity of 1000ft/minute or less. A meter reading shall be taken exactly five seconds after the element is subjected to the new medium. Another meter reading shall be taken exactly 60 seconds after the "zero time". A current ratio figure shall then be calculated from the current readings (I_0, and I_5 and I_{60} in accordance with the following:

The ratio $\frac{I_0 - I_5}{I_0 - I_{60}}$ must not be less than 0.90; where

I_0 = Current (in microamperes) at zero time
I_5 = Current after 5 seconds
I_{60} = Current after 60 seconds in order to satisfy the requirements of 3.9.8.

The ratio 0.90 is equivalent to a resistance change of 63 percent of the total. The procedure for -30°C time constant test shall be the same as that for the +25°C time constant test; except that in place of I_5 use I_2 (current after 2 minutes) and in place of I_{60} use I_{10} (current after ten minutes. (See 3.9.8).

4.9 Service conditions. The service condition tests shall be performed as detailed below. Each sensor subjected to these tests shall have passed group A tests and shall be resubmitted to and pass these group A tests after completion of all service condition tests.

4.9.1 Temperature test. The sensor sealed in its container shall be stored at a temperature of -60°C ± 2°C for a period of 48 hours; then at +60°C ±2°C for a period of 48 hours; and then conditioned in the container for 48 hours at 25°C ±2°C. Upon completion, it shall meet full specification requirements of 3.9.

4.9.2 Humidity test (container). The sensor sealed in its container shall be subjected to the temperature/humidity conditions of 3.10.1 followed by the pressure conditions specified in 3.10.2. Lack of leaks shall indicate isolation of the hygeoster from environmental effects.

4.9.3 Altitude test (container). The sensor in its container as prepared for shipment shall be subjected to an atmospheric pressure of 125 ± 5 millibars for thirty minutes. While the pressure is maintained there shall be no evidence of container deterioration or leakage.

4.9.4 Vibration. The sensor in its sealed container shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.10.3, as follows:

(a) The sensor in its normal shipping container shall be fastened on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinuousidal vibration.

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second in 1 cycle per second steps. The total excursion shall be constant at 1/64 inch at all frequencies.

4.10 Aging. Manufacturer's certified test data in conformance with the aging requirements of 3.11 will be acceptable as evidence of compliance.

4.11 Visual and mechanical inspection. The finished sensor and its container shall be visually checked for compliance with specific mechanical requirements and for defects listed in MIL-STD-252. (See 3.12). Particular attention shall be given to the following:

(a) The sealed container shall be examined for cracks, faulty sealing and proper identification marking.

(b) The humidity element shall be checked for continuity of electrodes and correctness of physical characteristics. (See 3.2 and 3.2.2).

(c) The electrodes shall be tested for bonding to the element by rubbing each surface of each electrode once longitudially with aclean soft cotton cloth, using firm finger pressure. The electrode shall show no evidence of peeling or removal. (See 3.2.1). Visual and mechanical tests shall be classified by the government as major or minor in accordance with definitions of MIL-STD-105.

(d) Classification of defects per Table V.

Table V Classification of defects
(visual and mechanical)

Major	Minor
Sealed container:	Illegible marking or not easily discernible from outside of container
Cracks	
Poor seal	
Improper marking	
Marking label missing	Missing dipping date.
Humidity element:	
Cracks in film	Warpage over 1/16 inch but less than 3/32 inch.
Cracks in electrodes	
Warpage over 3/32 inch	
Improperly cleaned	
Electrodes not securely bonded to element	
Dimensions outside of specified tolerances	
Weight in excess of specified limits	

4.12 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.12.1 First article inspection of packaging. Unless otherwise specified in the contract, First article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.12.1 First article inspection of packaging. Unless otherwise specified in the contract, First article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.12.2 Quality conformance inspection of packaging.

4.12.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.12.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.12.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table VI, Packing inspection provisions. Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABE VI. Packing inspection provision.

No.	CHARACTERISTIC	Method of inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1Ø1-1ØØ-OO-EA-X-B7-Q-ED-A.

5.1.2 Level B. 1Ø-1-1-ØØ-/OO/-EA-X-B7-Q-ED-B.

REQUIREMENT LESSONS LEARNED

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The Humidity Sensor, Resistance is used as a humidity sensor with Radisonde Set ML-659(V).

6.2 Ordering data. Procurement documents should specify the following

a. Title, number and date of this specification and any amendment thereto.
b. Level A or B preservation and packing (see section 5).
c. When first artticle inspection rough handling test(s) are not required.
d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.
e. Marking and shipping of samples.
f. Place of final inspection.
g. Number of first articles to be submitted for approval (see 3.1).
h. When first artticle packing inspection test reports require Acquisition Activity approval prior to production unit packing.

6.3 Level B preservation. When level B preservation is specified, this level of protectin will only be used under known favorable conditions during transportation, storage, and handling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.5 Subject terms (key word list)
Humitidy
Relative Humidity (RH)
Sensor
Humidity Sensor
Resistance Humidity Sensor
Resistance material plastic plate-hygroscopic coating
Radiosonde Set ML-659(V)

Custodian:
Army - CR

Preparing Activity
Army - CR
Project 6685 - A741

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-49448(CR)	Humidity Sensor Resistance

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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